

Отчёт по лабораторной работе 5

Простые сети в GNS3. Анализ трафика

Талебу Тенке Франк Устон

Российский университет дружбы народов, Москва, Россия

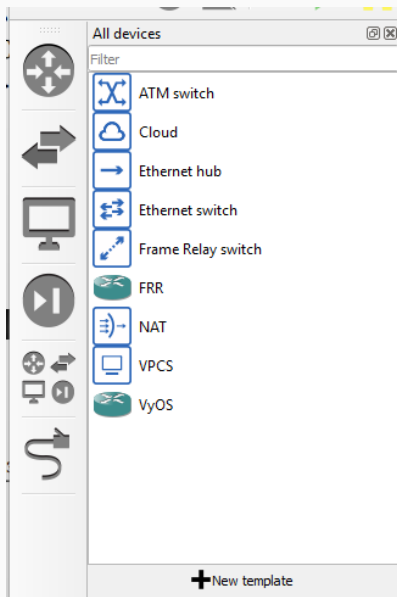
Цели и задачи работы

Цель лабораторной работы

Построение простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.

Выполнение лабораторной работы

Справка по командам VPCS



Справка по настройке IP

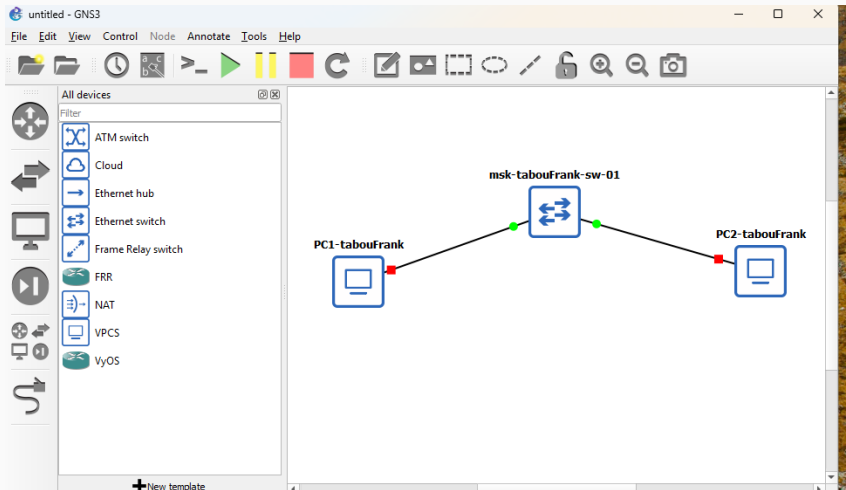


Рис. 2: Просмотр синтаксиса команды ip

Настройка IP-адреса PC1

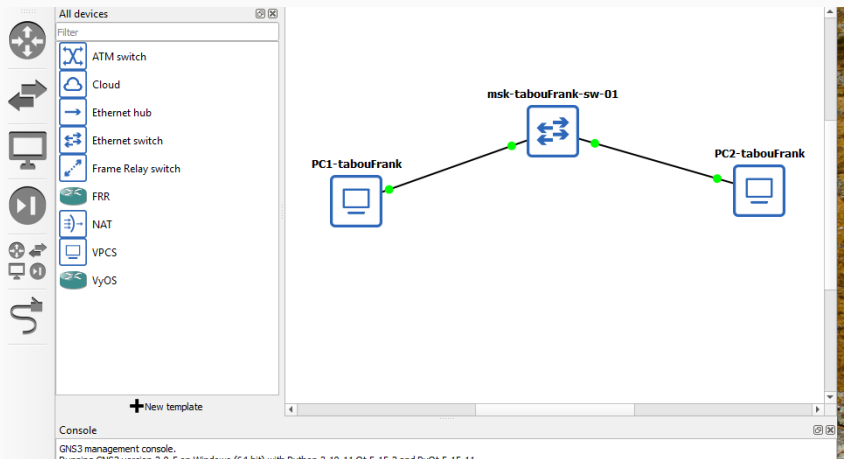
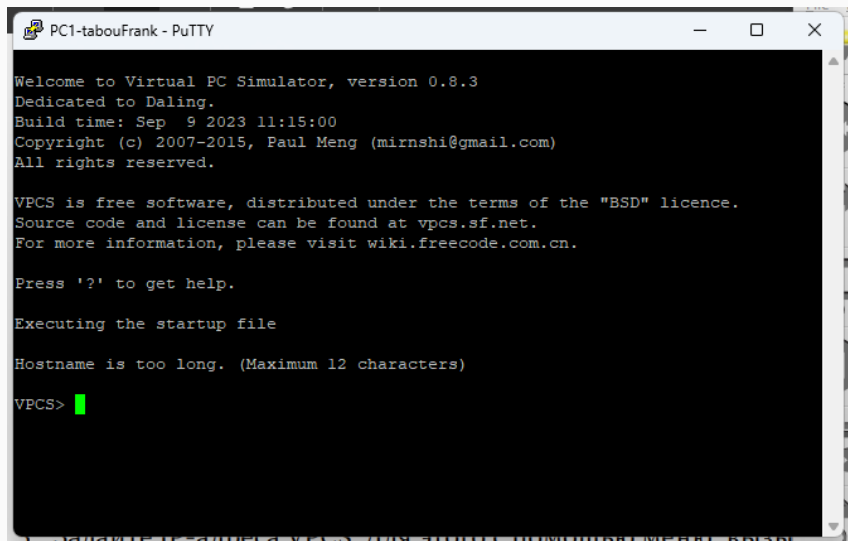


Рис. 3: Настройка IP-адреса PC1 и проверка параметров

Проверка связности PC1 ↔ PC2



```
PC1-tabouFrank - PuTTY

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep 9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> █
```

Рис. 4: Проверка связи между PC1 и PC2

ARP-сообщения (gratuitous ARP)

```
VPCS> /?

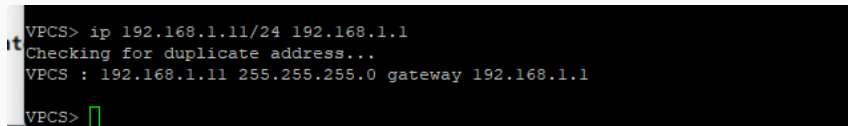
?                Print help
arp              Shortcut for: show arp. Show arp table
clear ARG        Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION]    Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect       Exit the telnet session (daemon mode)
echo TEXT        Display TEXT in output. See also set echo ?
help            Print help
history          Shortcut for: show history. List the command history
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?
load [FILENAME]  Load the configuration/script from the file FILENAME
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit            Quit program
relay ARG ...    Configure packet relay between UDP ports. See relay ?
rlogin [ip] port Telnet to port on host at ip (relative to host PC)
save [FILENAME]  Save the configuration to the file FILENAME
set ARG ...      Set VPC name and other options. Try set ?
show [ARG ...]   Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version          Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.

VPCS> █
```

Рис. 5: Анализ ARP-сообщений в Wireshark

Опции ping в VPCS



```
VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1
VPCS> 
```

Рис. 6: Просмотр параметров команды ping

Выполнение ping с PC2

```
VPCS> save
Saving startup configuration to startup.vpc
. done
VPCS> 
```

Рис. 7: Выполнение ping

ICMP Echo Request/Reply

```
VPCS> /?
/?
Print help
arp
Shortcut for: show arp. Show arp table
clear ARG
Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION]
Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect
Exit the telnet session (daemon mode)
echo TEXT
Display TEXT in output. See also set echo ?
help
Print help
history
Shortcut for: show history. List the command history
ip ARG ... [OPTION]
Configure the current VPC's IP settings. See ip ?
load [FILENAME]
Load the configuration/script from the file FILENAME
ping HOST [OPTION ...]
Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit
Quit program
relay ARG ...
Configure packet relay between UDP ports. See relay ?
rlogin [ip] port
Telnet to port on host at ip (relative to host PC)
save [FILENAME]
Save the configuration to the file FILENAME
set ARG ...
Set VPC name and other options. Try set ?
show [ARG ...]
Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT]
Print TEXT and pause running script for seconds
trace HOST [OPTION ...]
Print the path packets take to network HOST
version
Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.
VPCS> 
```

Рис. 8: Анализ ICMP Echo-запроса и ответа

UDP mode (Echo over UDP)

```
VPCS> /?

?          Print help
arp        Shortcut for: show arp. Show arp table
clear ARG  Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION] Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect Exit the telnet session (daemon mode)
echo TEXT  Display TEXT in output. See also set echo ?
help       Print help
history    Shortcut for: show history. List the command history
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?
load [FILENAME] Load the configuration/script from the file FILENAME
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit       Quit program
relay ARG ... Configure packet relay between UDP ports. See relay ?
rlogin [ip] port Telnet to port on host at ip (relative to host PC)
save [FILENAME] Save the configuration to the file FILENAME
set ARG ... Set VPC name and other options. Try set ?
show [ARG ...] Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version    Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.

VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

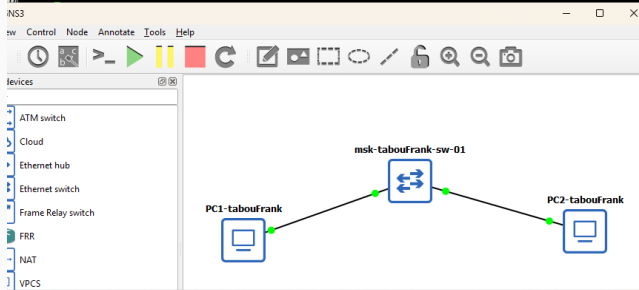
VPCS> 
```

TCP mode (Echo over TCP)

```
VPCS> ping

ping HOST [OPTION ...]
np Ping the network HOST. HOST can be an ip address or name
Options:
-1      ICMP mode, default
-2      UDP mode
-3      TCP mode
-c count Packet count, default 5
-D      Set the Don't Fragment bit
-f FLAG  Tcp header FLAG |C|E|U|A|P|R|S|F|
        bits |7 6 5 4 3 2 1 0|
-i ms    Wait ms milliseconds between sending each packet
-l size  Data size
-P protocol Use IP protocol in ping packets
        1 - ICMP (default), 17 - UDP, 6 - TCP
-p port  Destination port
-s port  Source port
-T ttl   Set ttl, default 64
-t       Send packets until interrupted by Ctrl+C
-w ms    Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.
```

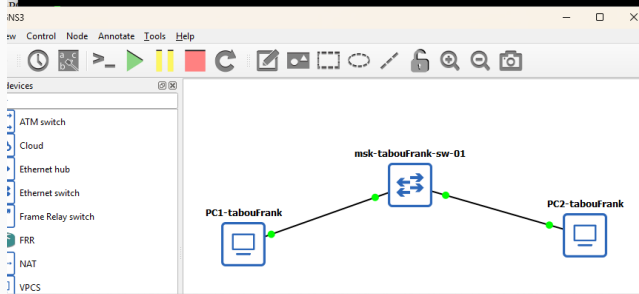


Топология: VPCS — Switch — FRR

```
VPCS> ping

ping HOST [OPTION ...]
pr Ping the network HOST. HOST can be an ip address or name
Options:
-1      ICMP mode, default
-2      UDP mode
-3      TCP mode
-c count Packet count, default 5
-D      Set the Don't Fragment bit
-f FLAG  Tcp header FLAG |C|E|U|A|P|R|S|F|
        bits |7 6 5 4 3 2 1 0|
-i ms    Wait ms milliseconds between sending each packet
-l size  Data size
-P protocol Use IP protocol in ping packets
        1 - ICMP (default), 17 - UDP, 6 - TCP
-p port  Destination port
-s port  Source port
-T ttl   Set ttl, default 64
-t       Send packets until interrupted by Ctrl+C
-w ms    Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.
```



Настройка IP на PC1 (192.168.1.10/24)

```
VPCS> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.815 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=2.968 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.948 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.397 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.327 ms

VPCS> 
```

Рис. 12: Настройка IP-адреса PC1 и проверка параметров

Настройка eth0 на FRR (192.168.1.1/24)

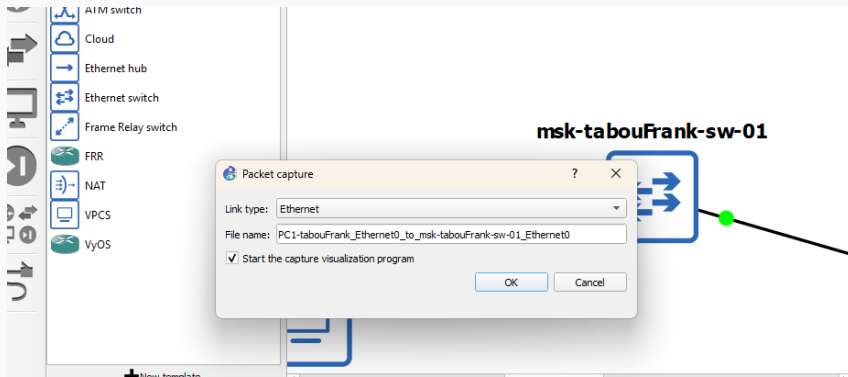
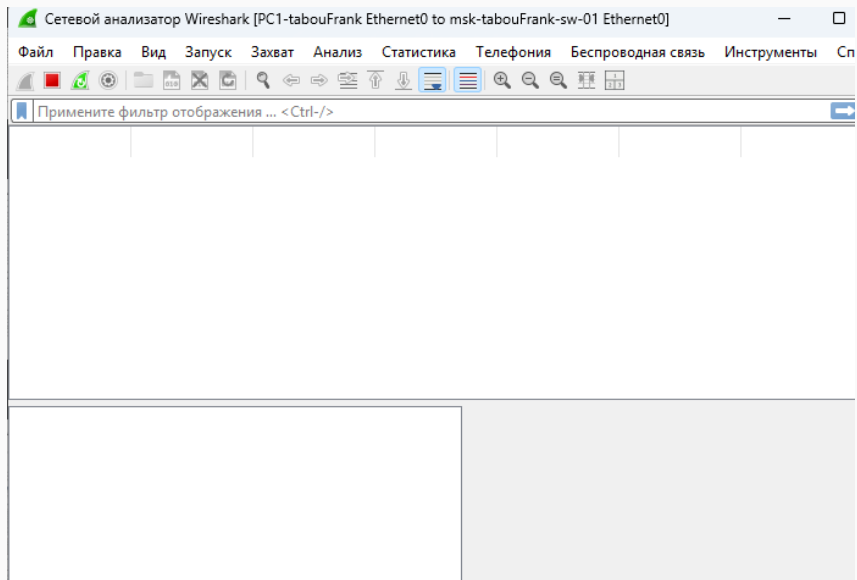


Рис. 13: Настройка интерфейса eth0 маршрутизатора FRR

Проверка конфигурации FRR



Проверка связности PC1 → FRR (192.168.1.1)

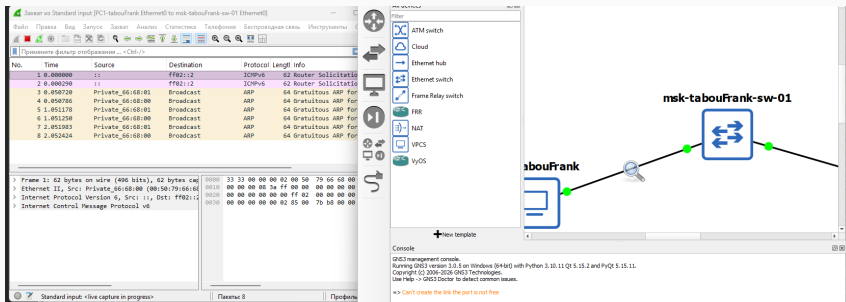


Рис. 15: Проверка связи между PC1 и маршрутизатором

ICMP/ARP в Wireshark (PC1 ↔ FRR)

The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes options like File, Edit, View, Capture, Analyze, Statistics, Telephony, Wireless, Instruments, and Help. Below the menu is a toolbar with various icons for file operations, capture control, analysis, and viewing. A filter bar at the top displays "Примените фильтр отображения ... <Ctrl-/>".

The main packet list pane shows eight captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.000290	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.050720	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
4	0.050786	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.051178	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.051250	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.051983	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.052424	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)

The bottom pane shows the detailed view of Frame 1 (selected). It indicates the frame is 62 bytes on wire (496 bits) and 62 bytes captured (496 bits) on interface 0 (-).

- Section number: 1
- Interface id: 0 (-)
- Encapsulation type: Ethernet (1)
- Arrival Time: Jan 2, 2026 21:09:11.089457000 RTZ 2 (sumo)
- UTC Arrival Time: Jan 2, 2026 18:09:11.089457000 UTC
- Epoch Arrival Time: 1767377351.089457000
- [Time shift for this packet: 0.00000000 seconds]
- [Time delta from previous captured frame: 0.00000000 seconds]
- [Time delta from previous displayed frame: 0.00000000 seconds]
- [Time since reference or first frame: 0.00000000 seconds]
- Frame Number: 1
- Frame Length: 62 bytes (496 bits)
- Capture Length: 62 bytes (496 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ipv6:icmpv6]
- [Coloring Rule Name: ICMP]
- [Coloring Rule String: icmp || icmpv6]

To the right of the details pane is a hex dump of the frame data, showing hexadecimal values and their corresponding ASCII representations.

Моделирование сети на базе VyOS

Топология: VPCS — Switch — VyOS

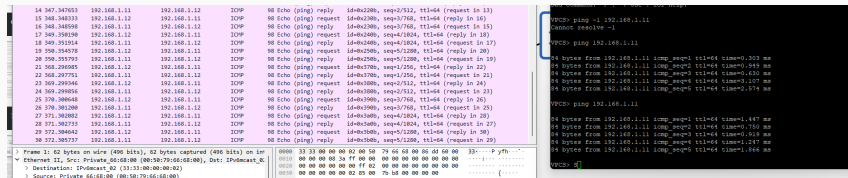
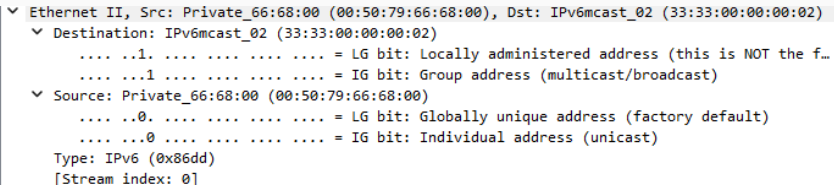


Рис. 17: Топология сети с маршрутизатором VyOS

Настройка интерфейса eth0 на VyOS (192.168.1.1/24)



▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: IPv6mcast_02 (33:33:00:00:00:02)

- ▼ Destination: IPv6mcast_02 (33:33:00:00:00:02)
 -1. = LG bit: Locally administered address (this is NOT the f...
 -1 = IG bit: Group address (multicast/broadcast)
- ▼ Source: Private_66:68:00 (00:50:79:66:68:00)
 -0. = LG bit: Globally unique address (factory default)
 -0 = IG bit: Individual address (unicast)

Type: IPv6 (0x86dd)
[Stream index: 0]

Рис. 18: Настройка интерфейса eth0 маршрутизатора VyOS

Проверка связности PC1 → VyOS (192.168.1.1)

```
.... ..0. .... = LG bit: Globally unique address (factory default)
.... ..0 .... = IG bit: Individual address (unicast)
▼ Source: Private_66:68:01 (00:50:79:66:68:01)
.... ..0. .... = LG bit: Globally unique address (factory default)
.... ..0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 4]
▼ Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 84
Identification: 0xcc5 (3269)
> 000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xea6c [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.12
Destination Address: 192.168.1.11
[Stream index: 0]
▼ User Datagram Protocol, Src Port: 6310, Dst Port: 7
Source Port: 6310
Destination Port: 7
Length: 64
Checksum: 0xc7ce [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 9]
> [Timestamp]
```

ICMP/ARP в Wireshark (PC1 ↔ VyOS)

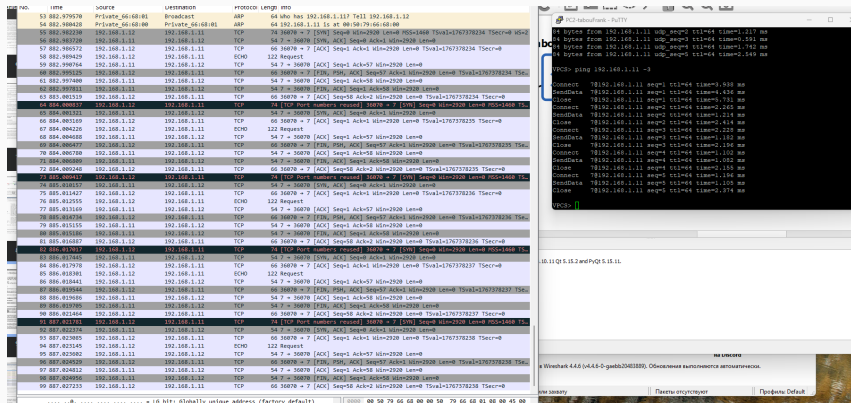


Рис. 20: Анализ ICMP-трафика между PC1 и маршрутизатором

Выводы по работе

В ходе выполнения лабораторной работы были смоделированы простейшие сети в GNS3 на базе коммутатора Ethernet и маршрутизаторов FRR и VyOS. Выполнена статическая настройка IP-адресации в подсети 192.168.1.0/24 и проверена связность между узлами с использованием ICMP эхо-запросов.

С помощью Wireshark были захвачены и проанализированы ARP- и ICMP-сообщения, а также особенности формирования трафика при выполнении ping в режимах ICMP/UDP/TCP. Поставленные задачи выполнены в полном объёме.